Zee Recommendation

Aug 01, 2024

→ Collaborative Filtering

→ Content based.

User-based Collaborative Alg.

Users	Matrix	Toy Story	Inception
Alice	4	5	?
Bob	5	3	4
Carol	4	?	4
Dave	?	3	5

Euclidian ✗

Sim(Alice, Bob) = 0.5

Sim(Alice, Carol) = 0.7

Sim(Alice, Dave) = 0.1 ✗ — Ignore

$$\text{Rating}_{\text{Alice, Inception}} = \frac{0.5 \times 4 + 0.7 \times 4}{0.5 + 0.7}$$

$$= 4$$

Item based collaborative Filtering Algorithm

Users	Matrix	Toy Story	Inception
Alice	4	5	?
Bob	5	3	4
Carol	4	?	4
Dave	?	3	5

Sim - Pearson

Sim (Inception, Toystory) = 0.1
" (Inception, Matrix) = 0.4

$$\text{Alice\_Inception} = \frac{0.9 \times 4 + 0.1 \times 5}{0.9 + 0.1}$$

$= \underline{4.1}$

Content - based Collaborative Algo

Movie	Genre	Director
Matrix	Sci-Fi	Lana Wachowski
Toy Story	Animation	John Lasseter
Inception	Sci-Fi	Christopher Nolan

Alice:

Matrix	Toy Story	Inception
5	3	4

Step - 1 Feature Vector

Movie	Sci-Fi (1)	Animation (2)	Directed by Lana Wachowski (3)	Directed by John Lasseter (4)	Directed by Christopher Nolan (5)
Matrix	1	0	1	0	0
Toy Story	0	1	0	1	0
Inception	1	0	0	0	1

Step - 2 Compute User - profile Vector

Matrix: $[1\ 0\ 1\ 0\ 0] \times 5 = [5, 0, 5, 0, 0]$

Toy story: $[0\ 1\ 0\ 1\ 0] \times 3 = [0, 3, 0, 3, 0]$

Inception: $[1\ 0\ 0\ 0\ 1] \times 4 = [4, 0, 0, 0, 4]$

User-Profile $= \left[\frac{5+0+4}{12}, \frac{0+3+0}{12}, \frac{5+0+0}{12}, \frac{0+3+0}{12} \right.$

$$\frac{0+0+4}{12}$$

$$= \begin{bmatrix} 0.75 & 0.25 & 0.42 & 0.25 & 0.33 \end{bmatrix}$$

Movie. $[1, 0, 0, 0, 1]$

Sim. $0.75 + \ldots + 0.33 = \underline{1.08}$

Precision@12

Metric:

1. MRR → mean reciprocal Rank.
2. NDCG → normalize discounted cumulative gain.

MRR → Good example

Q_1: Relevant result at $R\text{-}1 = \frac{1}{1}$

Q_2: " " " $R\text{-}2 = \frac{1}{2}$

Q_3: " " " $R\text{-}1 = \frac{1}{1}$

$$\frac{1 + 0.5 + 1}{3} = \underline{0.833}$$

Bad Example

Q_1 : Relevant hotel @ R-10 $= \frac{1}{10}$

Q_2 : " " @ R-5 $= \frac{1}{5}$

Q_3 : " " @ R-8 $= 1/8$

$$\frac{0.1 + 0.2 + 0.125}{3} = \underline{0.142}$$

CF / Content-based / →

xgboost / Lightgbm.

What is used in organizations like Amazon/Meesho:

L2R - Learning 2 Rank algorithms:

1. pair-wise ranking
2. list-wise ranking
3. point-wise ranking

Collaborati	Content-b
Cold-start	✗
No- Domain K	✓
Affecte- data sparsity	✗
Popular items -	✗
Affected - user pr	✗
Struggles with new items	✗

Matrix Factorization